COGNITIVE-LOAD-AWARE LEARNING SUPPORT: DESIGN, IMPLEMENTATION AND EVALUATION OF AN INDIVIDUAL COMPONENT

INDIVIDUAL THESIS — WORKING DRAFT

Dilshan W.G.W

IT22055712

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Supervisor: Dr Kalpani Manathunga

Co-supervisor: Jeewaka Sir

Sri Lanka Institute of Information Technology

Sri Lanka

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Student name: Dilshan W.G.W

Student ID: IT22055712

Signature: _____ Date: _____

Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under the supervision of the academic staff listed below.

Role	Name	Signature	Date
Supervisor	Dr Kalpani Manathunga	_____	_____
Co-supervisor	Jeewaka Sir	_____	_____

ABSTRACT

Digital learning platforms can record learner interactions, but model labels alone offer limited guidance to teachers and learners. This individual contribution to the Lumora learning platform includes two distinct teacher-facing explanation paths: a cognitive-style explanation that uses LIME and SHAP to identify influential visual-engagement observations behind a Visual, Verbal, or Intermediate classification, and a cognitive-load explanation grounded in positive behavioural contributions supporting an elevated load prediction. The component also provides learner recommendations, class-level next-lesson guidance, and adaptive multiple-choice questions (MCQs). Structured generative-AI outputs are subject to evidence, schema, and workflow checks; individual guidance and next-lesson recommendations include teacher review, while MCQ structural validation does not establish answer-key correctness. Automated testing recorded 39 passing tests across the project-wide guidance, recommendation, and MCQ groups, with a separate cognitive-style service suite passing 10 tests. Preliminary blinded evaluations by three ICT teachers rated the load-related system guidance at 4.84/5 compared with 3.87/5 for baseline guidance and preferred it in nine of nine scenario comparisons. These ratings did not separately evaluate the cognitive-style explanation. Adaptive MCQs received higher descriptive ratings than baseline MCQs; teacher answer-key review marked 29/30 adaptive answers and 30/30 baseline answers as correct and uniquely valid. A separate classroom pilot involved ten Grade 8 learners who studied the INTERNET lesson on the hosted site and completed a generated ten-question quiz; 78/100 answers were correct (78.0%). Learners also rated MCQ clarity on a 1–5 scale, and one teacher reviewed the MCQs and next-lesson recommendation. These small, descriptive records do not establish learning gains, broad usability, style-explanation quality, or general correctness.

Keywords: cognitive load, explainable AI, LIME, SHAP, teacher guidance, adaptive MCQs

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LIST OF ABBREVIATIONS

AI — Artificial Intelligence
API — Application Programming Interface
CoVe — Chain-of-Verification
LLM — Large Language Model
LIME — Local Interpretable Model-agnostic Explanations
MCQ — Multiple-Choice Question
SHAP — SHapley Additive exPlanations
XAI — Explainable Artificial Intelligence

1 INTRODUCTION

1.1 Overall project context

The project develops a cognitive-load-aware learning platform intended to connect digital learning activity with support for teachers and learners. Video lessons create observable interaction traces such as pauses, seeking, replaying, playback-speed changes, inactivity, and time spent with content. A prediction service maps these observations to a cognitive-load category, while a separate cognitive-style model classifies a learner as Visual, Verbal, or Intermediate from visual-engagement observations. This thesis covers the individual contribution's explanation of both model outputs, as well as learner recommendations, next-lesson guidance, and assessment questions. It does not claim ownership of the upstream cognitive-load model or the collection and training of the cognitive-style classifier.

The platform's cognitive-load output is a model estimate for a lesson context. It is not a direct measurement of a learner's private mental state and is not a diagnosis, ability score, or permanent learner label. The behavioural indicators are ambiguous: for example, replaying a segment may reflect difficulty, careful review, interruption, or interest. Therefore, the component uses cautious language and keeps the source evidence inspectable.

1.2 Individual component context

The individual component has five primary outputs. First, it creates a teacher-facing explanation of a student's predicted cognitive style using style-specific LIME and SHAP evidence. Second, it creates a separate teacher-facing explanation of a student's lesson-level cognitive-load estimate and provides four learner-facing actions with a constrained study-technique suggestion. Third, it creates a class-level recommendation for the teacher's next lesson from structured course evidence. Fourth, it generates ten MCQs from lecture material, adapting question wording and demand to a cognitive-load category. Style analysis explains why the classifier selected a style; it does not provide recommendations or teaching actions. Individual guidance and next-lesson recommendations support teacher review; the MCQ service checks structure but does not automatically establish semantic answer correctness.

The work combines explainability, deterministic processing, generative AI, service integration, and human review. It is an engineering and applied-evaluation contribution rather than a claim to introduce a new cognitive-load theory or foundation model.

1.3 Problem statement

A category such as High or Very High cognitive load, or a Visual/Verbal/Intermediate cognitive-style label, is difficult to interpret if teachers cannot inspect the observations that influenced it. Raw feature names and LIME/SHAP values are unsuitable as the only explanation for routine classroom decisions. Generic learner advice may not connect to observed behaviour, and class recommendations can overstate what limited evidence supports. Generative systems can produce plausible but unsupported explanations, recommendations, questions, or answer keys. A practical system therefore needs to translate the two distinct model outputs into understandable, evidence-bounded explanations while preserving structural checks, teacher oversight, and transparent limitations.

1.4 Research gap

The project documents a gap between a model output and a usable instructional response: users need a traceable route from predicted load to understandable explanation, learner action, class planning, and lecture-grounded assessment. The component addresses this integration gap by combining local explanation signals, controlled language generation, deterministic validation, evidence snapshots, and review workflows. The current evaluation is preliminary. The available small teacher studies and classroom worksheet do not establish that the integrated approach improves learning or generalises to other subjects and populations.

1.5 Aim and objectives

Aim: To design, implement, and preliminarily evaluate a traceable cognitive-load-aware support component that helps teachers and learners interpret lesson evidence and use generated educational support.

Objectives:

1. Combine cognitive-style LIME and SHAP attributions by normalised magnitude and retain their direction when selecting influential observations.

2. Generate a cautious, teacher-friendly explanation of the predicted cognitive style from the selected style evidence.

3. Combine positive cognitive-load LIME and SHAP contributions and use them to generate load-related teacher explanation and learner recommendations.

4. Aggregate class-level evidence to generate a reviewable next-lesson recommendation with verification and fallback behaviour.

5. Generate lecture-grounded MCQs adapted to a cognitive-load category and validate their output structure.

6. Evaluate software behaviour and report available teacher/learner feedback accurately, distinguishing load-guidance evaluation from cognitive-style explanation evaluation.

1.6 Research questions

RQ1: How can local LIME and SHAP evidence for cognitive-style and cognitive-load predictions be combined and presented as cautious, teacher-readable explanations? **RQ2:** How do teachers descriptively rate system-generated guidance compared with baseline guidance? **RQ3:** How do teachers descriptively judge adaptive lecture-grounded MCQs compared with baseline MCQs? **RQ4:** What preliminary evidence about quiz performance, perceived clarity, teacher review, and recommendation suitability is recorded in the exploratory classroom user-test workbook?

RQ4 is exploratory and limited to the entries currently present in the workbook. It is not a controlled learning-effectiveness question.

1.7 Scope, limitations, and assumptions

This thesis covers teacher-facing cognitive-style explanation and cognitive-load explanation, learner recommendations, class-level next-lesson recommendation, and

MCQ generation. Study-technique selection is discussed as part of the load-guidance bundle but is not a separate primary outcome. The upstream cognitive-load classifier and the capture/training workflow for the cognitive-style classifier are dependencies rather than contributions claimed here. The cognitive-style explanation service consumes the classifier output and saved visual-engagement inputs; the load-guidance and assessment services consume the lesson-level cognitive-load record and extracted lecture text.

The component does not claim that behavioural traces reveal causes, that the selected positive signals show every factor behind a prediction, that generated questions are always correct, or that one quiz score demonstrates improved learning. The classroom workbook has ten learner rows and one teacher-review entry; this small sample does not support broad effectiveness or generalisation claims.

The cognitive-style explanation is implemented as a separate teacher-facing XAI workflow. It explains the classifier's Visual, Verbal, or Intermediate output using local visual-engagement evidence; it does not adapt lecture content, diagnose ability, or generate instructional recommendations. The available teacher A/B evaluation workbook assesses cognitive-load guidance and MCQs, not this separate style explanation. Consequently, the style explanation is documented as an implemented function but has no independent teacher-quality or learner-outcome result in the reported evaluation.

1.8 Individual contribution

The individual contribution comprises two explainability flows: (a) cognitive-style LIME/SHAP aggregation and a teacher-friendly explanation of the predicted style, and (b) positive cognitive-load evidence ranking followed by teacher explanation and learner recommendations. It also includes constrained study-technique selection, class-level next-lesson planning with a CoVe verification flow, lecture-grounded MCQ generation, integration with platform services, functional testing, and preparation/analysis of the available evaluation artefacts. The upstream cognitive-load prediction model, cognitive-style data collection/classifier training, course environment, and other students' components are outside the contribution claimed

here. This scope reflects the component boundaries documented in the project repository.

1.9 Thesis structure

Chapter 1 introduces the problem and contribution. Chapter 2 reviews cognitive load, explainability, adaptive support, question generation, and human oversight. Chapter 3 describes the evaluation methods and measures. Chapter 4 presents requirements and system design. Chapter 5 documents implementation, integration, and software testing. Chapter 6 reports the existing teacher evaluations and the exploratory classroom worksheet. Chapter 7 interprets the findings and limitations. Chapter 8 considers practical value and sustainability. Chapter 9 concludes and identifies future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review focuses on sources directly related to cognitive load, local feature explanations, cognitive-style interpretation, feedback and retrieval practice, generative AI in education, and verification of model-generated claims. It is a narrative, project-scoped review rather than a systematic literature review; no exhaustive database search, screening protocol, or formal quality appraisal was conducted.

2.2 Domain and theoretical foundations

2.2.1 Cognitive load in digital learning

Cognitive Load Theory describes the constraints imposed by working memory during learning and problem solving [1]. Instructional support can be adjusted to task demands, but the platform's interaction features remain proxies rather than direct measurements of mental effort. Pausing, replaying, seeking, changing playback speed, inactivity, and time on content each admit multiple interpretations. This thesis therefore treats the predicted category as contextual evidence for a support decision, not a fixed property of a learner.

2.2.2 Local explainability

LIME approximates a model's local decision with an interpretable surrogate [2], while SHAP assigns additive feature attributions based on Shapley-value concepts [3]. Their scales and assumptions differ, and their outputs are explanations of model behaviour rather than causal explanations of cognition. The component defines a transparent engineering aggregation rule: exclude non-positive values for the specific purpose of showing signals supporting increased predicted load, normalise each explainer by its own positive total, combine the shares equally, and rank the resulting values. This is a component-specific policy, not a universal XAI consensus algorithm.

2.2.3 Cognitive style and explanation boundaries

In this project, cognitive style is a model classification based on visual-engagement observations, with Visual, Verbal, and Intermediate outputs. It is treated as a contextual profile, not a fixed learner trait or a measure of ability. Local LIME and SHAP attributions can help a teacher inspect which inputs influenced a particular class prediction, but they explain the classifier's behaviour rather than establish a causal account of how a learner thinks. Accordingly, the generated explanation translates selected observations into cautious classroom language and deliberately excludes recommendations or teaching actions.

2.2.4 Feedback and retrieval activities

Feedback can support learning when it gives useful information about a task and next steps [4]. Retrieval practice can also support retention and learning [5]. These findings motivate short self-check questions and actionable recommendations, but they do not imply that every generated quiz or recommendation will be effective. The educational value of an item depends on its accuracy, alignment, difficulty, and use in context.

2.2.5 Generative AI and verification

Generative language models can produce fluent explanations, suggestions, and assessment drafts, but fluency does not ensure factual support or pedagogical suitability. Human-centred educational use therefore requires boundaries, review, and awareness of limitations [6]. Chain-of-Verification (CoVe) structures a response process as drafting, planning verification questions, checking those questions against evidence, and synthesising a grounded answer [7]. In this component, verification is performed against a structured evidence snapshot, not raw lecture files or source materials.

2.3 Related research

Research on explainable educational data mining motivates the use of local explanations to make predictions easier to inspect. Niu et al. compare explanation methods in an educational data-mining context [8]. Work comparing XAI methods

also highlights that different attribution methods can produce different outputs and should not be treated as interchangeable truth. In this implementation, LIME and SHAP are used as complementary ranking inputs; consensus is operationalised in code and evaluated as such.

The cognitive-style explanation is a second, separate use of local explainability. It ranks input observations by the mean of their separately normalised absolute LIME and SHAP magnitudes, while retaining the combined sign to describe whether each observation supports or opposes the selected class. Unlike the load-support list, it does not discard negative attributions: the purpose is to explain the selected style classification using the strongest influences in either direction. This ranking is an implementation decision and not evidence that the selected behaviours cause a learner's style.

Research on feedback, retrieval practice, educational question generation, and AI-supported learning motivates the learner recommendations and MCQ workflow. The design responds to known risks by grounding MCQ prompts in extracted lecture text and validating the generated schema. However, semantic alignment, uniqueness of the correct answer, and answer-key correctness require subject-matter review. A validator that confirms ten questions and four options cannot by itself verify the educational content.

2.4 Existing systems, products, and techniques

Learning analytics dashboards commonly expose indicators or risk categories. The component extends this pattern by connecting evidence to a teacher explanation and learner actions, and by deriving class-level lesson planning and quizzes. LIME and SHAP are established post-hoc explanation techniques; structured prompting, JSON parsing, fixed catalogues, deterministic fallbacks, teacher review, and CoVe are control techniques rather than novel algorithms. The contribution lies in their integration into a multi-output workflow with different evidence and review policies for each output.

2.5 Gap synthesis and conceptual framework

The conceptual chain has two distinct explanation branches: visual-engagement inputs → predicted cognitive-style class → LIME/SHAP attribution ranking → teacher-friendly style explanation; and lesson interactions → predicted cognitive-load category → positive LIME/SHAP evidence → teacher explanation and learner guidance. Class-level load and style summaries separately support next-lesson planning, while lecture text supplies the factual boundary for MCQ generation. Each transition can introduce uncertainty, so the interface and thesis preserve provenance and distinguish prediction, explanation, recommendation, and assessment. The current evaluations provide preliminary evidence for load guidance and MCQ perception, but do not separately evaluate style-explanation quality.

3 RESEARCH METHODOLOGY

3.1 Research design

The work follows a design-and-development approach with multiple evaluation layers. First, automated tests assess defined software behaviours, including a separate ten-test suite for the cognitive-style explainability service. Second, a preliminary blinded paired teacher evaluation compares baseline and system outputs for cognitive-load guidance and MCQs; it does not separately rate cognitive-style explanations. Third, an exploratory classroom session involved ten Grade 8 learners and one teacher, using a hosted INTERNET lesson, generated MCQs, learner clarity ratings, and teacher review. These are complementary forms of evidence and are not combined into a single effectiveness estimate.

3.2 Data, participants, and study materials

The teacher-guidance evaluation workbook records three teacher codes (T001–T003), three scenarios, ratings for two methods, and preferences. This yields nine paired scenario evaluations. The MCQ expert-evaluation workbook records three ICT teacher responses across baseline and adaptive question sets, including per-item answer-key judgments.

The classroom user-testing workbook records one INTERNET lesson session on 3 October 2026 with ten Grade 8 learners (S01–S10) and one teacher. Learners viewed the lesson on the hosted site, completed the MCQ quiz generated by the site, and the teacher marked their answers. Afterward, learners rated whether they understood the MCQs using a 1–5 choice and could leave a comment; no learner comments were entered. The teacher reviewed the MCQs and their answers, completed the review fields, and viewed the system's next-lesson recommendation. The workbook contains one completed teacher-review row. The exact wording and labels for each point on the learner rating scale were not recorded. This one-teacher classroom review is separate from the earlier blinded A/B evaluations involving three teachers.

Study materials include baseline and adaptive MCQ packs, lecture content, the generated quiz, system cognitive-load outputs, teacher guidance outputs, evaluation

forms, and the two evaluation spreadsheets. The classroom scores are treated as one post-lesson quiz outcome, not a pre/post measure.

3.3 Experimental and study procedure

3.3.1 Automated functional tests

The project-wide test report groups 39 checks across individual guidance and signal aggregation, next-lesson and CoVe workflows, MCQ generation, and lecture PDF extraction. Separately, the cognitive-style service test suite contains ten checks. Provider responses are mocked for deterministic tests where documented. These tests verify specified software behaviours rather than live language-model quality or learning outcomes.

3.3.2 Blinded teacher evaluations

For guidance, three ICT teachers compared baseline and system outputs for three scenarios. The Set A/Set B mapping was counterbalanced across scenarios, and ratings covered clarity, evidence relevance, recommendation usefulness, technique suitability, overall usefulness, and preference. For MCQs, three teachers evaluated a baseline set and an adaptive set for each of three lesson packs without being shown the method mapping before evaluation. They rated wording manageability, difficulty suitability, directness, overall suitability, and item answer correctness/uniqueness.

3.3.3 Exploratory classroom user test

The session took place on 3 October 2026 with ten Grade 8 learners and one teacher. Learners accessed the hosted site, viewed the INTERNET lesson, and completed the ten MCQs generated from that lesson. The teacher marked the answers; the recorded correct counts and percentages are in the data-collection workbook. After the quiz, each learner selected a 1–5 response about how well they understood the MCQs and had an opportunity to add a comment; no comments were submitted. The teacher then reviewed the MCQs and answers, completed the teacher evaluation fields, and viewed the system's next-lesson recommendation. The workbook records one teacher review. As the session log does not contain timings or detailed verbal observations, none are reported.

3.4 Measures and evaluation metrics

Automated evaluation uses test counts and pass/fail outcomes. Teacher guidance evaluation uses 1–5 ratings by criterion, overall method preference, and comments. MCQ evaluation uses criterion means, set preference, and the count of items rated as having a correct and uniquely valid system-selected answer. The classroom worksheet uses number correct out of ten and its derived percentage. It also records system cognitive-load category, learner clarity rating where populated, and a teacher's categorical review of content match, answer key, difficulty, wording, class understanding, recommendation suitability, and usefulness.

The learner 1–5 item concerns understanding/clarity of the MCQs; it is not a separate perceived-difficulty measure. The dataset contains no perceived-difficulty responses, so no such statistic is reported. Ratings and quiz scores are summarised descriptively; the small samples do not support inferential tests or population-level claims.

3.5 Data analysis

For the classroom worksheet, the ten recorded correct counts are summed and divided by 100 possible answers. The aggregate percentage and mean learner percentage are identical because every learner answered ten items. Clarity ratings are averaged over the ten available entries. Cognitive-load labels are counted as recorded. The single teacher row is reported as one person's evaluation, not as a consensus.

For paired teacher studies, repository summary values are reproduced as descriptive means and preference counts. The answer-key result is reported as affirmative judgments out of 30 per method. No significance tests are performed because the evaluators are few and the observations are clustered by evaluator and lesson. A rating difference is not interpreted as proof of causal superiority.

3.6 Ethics, privacy, security, and responsible AI

The classroom spreadsheet records participant codes rather than learner names, and this report presents no direct learner identifiers. Cognitive-load predictions are contextual estimates, not diagnoses or fixed learner labels. Generated explanations

and recommendations are decision-support outputs; teachers remain responsible for reviewing educational content before classroom use.

3.7 Reproducibility and research artefacts

The analysis draws on the individual thesis content template, formatting standards, project research paper, component workflow explanation, automated-test report, teacher-guidance evaluation workbook, MCQ expert-response workbook, classroom user-testing workbook, paper teacher-review form, implementation services, and document-generation scripts held in the project repository. Results are reported from those artefacts; no unrecorded observations or participant responses are inferred.

Table 3.1. Evaluation stages and evidence

Stage	Sample / cases recorded	Main evidence	Interpretation
Project-wide automated tests	39 tests	Pass/fail by component	Functional behaviour only
Cognitive-style service tests	10 tests	Pass/fail by service test suite	Functional behaviour only; no teacher-quality measure
Teacher guidance A/B	3 teachers × 3 scenarios	Ratings and preference	Preliminary expert perception
MCQ expert comparison	3 teachers × 3 lesson packs	Ratings and answer-key judgments	Preliminary content perception
Classroom worksheet	10 populated learners; 1 teacher row	Quiz scores, cognitive-load labels, categorical review	Exploratory descriptive evidence

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Table 4.1. Functional requirements

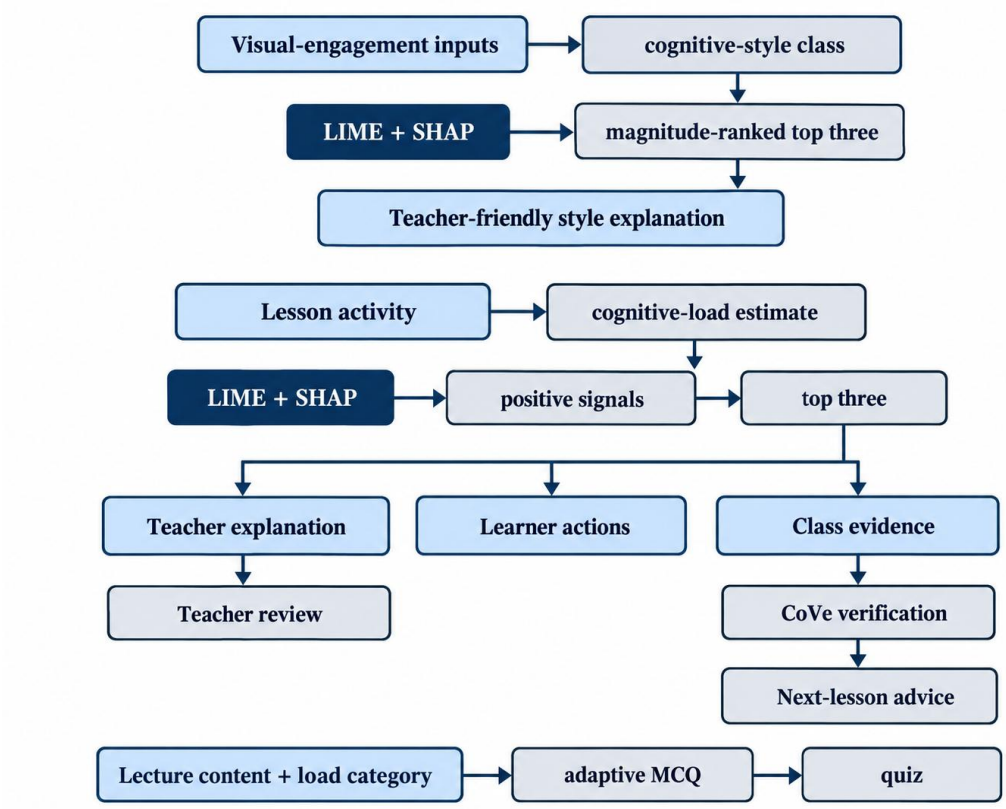
ID	Requirement	Intended outcome
FR1	Aggregate positive LIME and SHAP values	Rank up to three supporting signals
FR2	Generate a teacher-facing explanation	Explain the lesson-level estimate in cautious language
FR3	Generate learner recommendations	Provide four concise, actionable suggestions
FR4	Generate next-lesson guidance	Use class-level structured evidence with a fallback
FR5	Generate lecture-grounded MCQs	Return ten questions, four options, a key, and explanation
FR6	Support teacher review	Keep individual and class recommendations reviewable
FR7	Separate evidence from advice	Avoid presenting a recommendation as an observed fact
FR8	Explain cognitive-style predictions	Present the top influential style observations in cautious, teacher-friendly language

Non-functional requirements include traceability, authenticated access, evidence minimisation, deterministic handling of malformed output, and clear presentation of uncertainty.

4.2 Component architecture within the complete solution

The browser-based platform connects learning activity, cognitive-load inference, cognitive-style analysis, explanation services, recommendation generation, and assessment generation through authenticated APIs. The individual contribution includes a style explainability service that consumes the saved style-model inputs and predicted class, alongside the separate load-guidance service and the class recommendation and examination services. Outputs are stored in their respective records. Database and ownership checks remain service-side responsibilities.

Figure 4.1. Individual component workflow within the platform



4.3 Design alternatives and decisions

The load-explanation service is intended to list signals supporting an elevated load prediction. It therefore excludes non-positive values before normalisation, intentionally omitting load-reducing evidence from that purpose-specific list. This rule is not reused for cognitive-style explanations: style attribution ranks by absolute normalised importance and preserves each feature's direction, so influential evidence that opposes the selected style can also be described. Neither explanation presents its selected top three as a complete account of all model influences.

The generation path uses structured outputs with deterministic validation rather than unconstrained free text. Fixed technique catalogues reduce unsupported strategy names. For class recommendations, a deterministic fallback is retained if the multi-stage generation or verification flow fails. MCQ output is validated for shape and duplication, with expert review treated separately because structural validation cannot prove content correctness.

4.4 Detailed design

4.4.1 Positive LIME–SHAP aggregation

For feature *i*, the system retains positive contributions ($l_i^+ = \max(l_i, 0)$) and ($s_i^+ = \max(s_i, 0)$). Each explainer is normalised by its own positive total:

$$\text{LIME_share}_i = \frac{l_i^+}{\sum_j l_j^+}$$

$$\text{SHAP_share}_i = \frac{s_i^+}{\sum_j s_j^+}$$

$$\text{combined}_i = 0.5 \times \text{LIME_share}_i + 0.5 \times \text{SHAP_share}_i$$

An explainer with no positive total contributes zero. Equivalent LIME rules and SHAP names are mapped to canonical descriptors. Signals are sorted by descending combined score; canonical names break ties. Up to three signals are selected. Non-positive values are not included in this load-increasing support list; their exclusion is a deliberate filtering policy, not an accidental loss of data.

4.4.2 Cognitive-style explanation

The style service uses four model inputs: image cursor ratio, image scroll ratio, image gaze ratio, and whether the learner's first interaction was with visual content. The existing classifier predicts Visual, Verbal, or Intermediate; this contribution explains that output and does not claim to build or retrain the classifier. LIME and SHAP are run concurrently for the predicted class. Their outputs are independently normalised by the sum of absolute attribution magnitudes. For feature *i*, importance is the mean of the absolute normalised LIME and SHAP values, while direction is the sign of the mean signed normalised values. Features are ranked by importance and up to three are retained. Positive, negative, and neutral directions are preserved; unlike the load-support pipeline, negative style attributions are not filtered out.

Known feature names are translated into plain-language descriptions before prompting. A constrained Gemini prompt receives the predicted style, confidence, and ranked observations and requests one coherent 65–100-word explanation for a non-technical teacher. It asks for cautious language, prohibits raw technical terms and numerical values, and explicitly forbids advice, recommendations, diagnoses, or

permanent-trait claims. The generated explanation, prompt, model name, prediction, confidence, raw attributions, and selected features are stored with the analysis. The interface presents the teacher-friendly explanation by default and keeps technical evidence behind a separate control. This is a generated rationale for model behaviour, not a causal or psychological explanation.

4.4.3 Individual explanation and learner recommendations

The guidance request receives a cognitive-load category and selected behavioural phrases. A structured generation response provides a teacher explanation, exactly four learner recommendations, and one or two techniques from a fixed catalogue. Output parsing, phrase matching, technique checks, and list-size validation reject defined malformed responses. Some natural-language requirements remain prompt-based and are not fully enforced by deterministic code. Guidance begins in a pending state and teacher controls support approval, rejection, or regeneration before sharing.

4.4.4 Next-lesson recommendation

The recommendation service aggregates lesson-level class results, calculates category counts and high/very-high proportions, ranks recurring evidence signals, and composes a structured evidence snapshot. The CoVe sequence drafts a plan, generates verification questions, answers them against the snapshot, and synthesises a final recommendation. Valid evidence paths are required for supported claims; deterministic templates provide a fallback. The evidence snapshot is structured database evidence, not raw PDFs, slides, or video transcripts.

4.4.5 Adaptive MCQ generation

The MCQ service retrieves extracted lecture text and a trusted cognitive-load category, then requests ten questions with four options, a selected answer, and an explanation. Higher predicted load prompts concise and direct stems; medium load requests a balance of recall, comprehension, and simple application; lower load permits more application and inference while remaining within lecture content. The validator checks count, option count, duplicates, non-empty fields, and answer labels. It cannot determine whether the answer is factually correct or unique.

4.5 Standards and professional practice

The document uses the formatting conventions in the SLIIT dissertation guide supplied with the project, including Times New Roman, chapter and section heading hierarchies, 1.5-line body text, the stated page margins, preliminary Roman pagination, and Arabic numbering for the main text. Tables and figures are numbered and captioned. The table of contents and page numbering were regenerated after the content changes. Departmental submission requirements remain governed by the institution's current instructions.

5 IMPLEMENTATION, INTEGRATION, AND TESTING

5.1 Development environment and technology stack

The implementation uses a React-based frontend and service-oriented backend. The individual functions are implemented across Python and JavaScript services, with SQL and document databases used for different records and metadata. The cognitive-style explainability service uses FastAPI/Python with LIME and Kernel SHAP, persists analysis and generated explanation records in MySQL, and requests teacher-friendly text from Gemini. The load-guidance path uses a separate Python service; next-lesson planning and MCQ generation use JavaScript services. Exact dependency and deployment versions are not itemised in the available evaluation records, which limits exact environment reconstruction.

5.2 Component implementation

Table 5.1. Component responsibilities

Component	Responsibility
`lime_ai/services/prediction_service.py`	Signal aggregation, ranking, persistence, review state
`lime_ai/services/student_guidance_service.py`	Teacher explanation and learner guidance generation/validation
`cognitive_style_ai/services/explanation_service.py`	Cognitive-style LIME/SHAP execution and magnitude/direction aggregation
`cognitive_style_ai/services/human_explanation_service.py`	Plain-language cognitive-style explanation prompt and feature translation
`cognitive_style_ai/routers/api.py`	Style analysis workflow, persistence, cache, and API response
`recommendation_ai/server.js`	Class evidence aggregation, CoVe, next-lesson recommendation, review workflow
`exam_service/src/services/geminiMcq.js`	Adaptive MCQ prompt, structured response, local validation
`exam_service/src/routes/quizzes.js`	Lecture context retrieval, quiz persistence, response checking

The two explanation paths use different aggregation policies: cognitive-style explanation retains positive and negative feature directions while ranking by attribution magnitude; cognitive-load guidance selects only positive support for an

elevated-load list. The style interface avoids raw attribution terminology in its default teacher explanation and offers technical evidence separately. The next-lesson service limits its generation input to an aggregated evidence object. The MCQ path uses lecture text and a trusted load category and retains correct answers server-side during an attempt, returning results after submission. These service boundaries support task-specific data handling and validation.

5.3 Integration with the other components

The component depends on upstream cognitive-load predictions, the existing cognitive-style classifier and its saved behaviour inputs, course and learner records, and lecture uploads/extracted text. Authenticated APIs connect the browser to the services. Teacher ownership and learner enrolment are checked server-side for relevant data. The style explainer analyzes and explains the existing classifier output; it does not gather the original cursor/gaze behaviour or train that model. The recommendation uses course-level aggregates; MCQ generation uses the learner's course/lesson context. Failures in model generation or verification can trigger deterministic fallbacks for next-lesson guidance, while MCQ schema failure is rejected rather than treated as valid educational material.

5.4 Verification and software testing

The repository's project-wide report documents 39 passing automated tests: 20 for individual guidance and signal aggregation, 13 for next-lesson and CoVe logic, four for MCQ generation, and two for lecture PDF extraction. Separately, the cognitive-style service suite was run for this thesis revision and passed 10/10 tests covering attribution aggregation, teacher-prompt content and label handling, caching/model refresh, MongoDB input synchronization, and loading the three-class model artifact. Passing these tests supports specified software behaviours only. It does not establish that the style classifier is accurate, that generated explanations are pedagogically valid, or that teachers understand or trust them.

5.5 Deployment and operation

The classroom pilot used the hosted workflow, which depended on authentication, service connectivity, database records, lesson uploads, and model-provider

configuration. The user-testing workbook does not record a build identifier or exact provider configuration, limiting exact reproduction of the hosted session. Generated educational outputs remain decision-support material, and access to student records is limited by the platform's authenticated workflows. Production configuration and security controls depend on the deployment environment and are not evaluated by the classroom workbook.

6 RESULTS AND VALIDATION

6.1 Research results: teacher-guidance comparison

The teacher-guidance workbook contains nine paired scenario comparisons from three teachers. The completed summary reports that the system method was preferred in all nine comparisons. Averaged across the five reported rating criteria, the system mean was 4.84/5 and the baseline mean was 3.87/5. The separate overall-usefulness item was 5.00/5 for system guidance and 4.00/5 for baseline guidance. Criterion means were also higher for the system output in this dataset.

Table 6.1. Preliminary teacher-guidance ratings (n = 9 paired scenario ratings)

Criterion	Baseline mean (/5)	System mean (/5)	Difference
Clarity	3.44	4.89	+1.44
Evidence relevance	3.67	4.78	+1.11
Recommendation usefulness	4.33	4.78	+0.44
Technique suitability	3.89	4.78	+0.89
Overall usefulness	4.00	5.00	+1.00

Figure 6.1 visualises the criterion means, and Figure 6.2 shows the method preferences. The figures are descriptive summaries of three teachers rating three scenarios, not nine independent teacher participants.

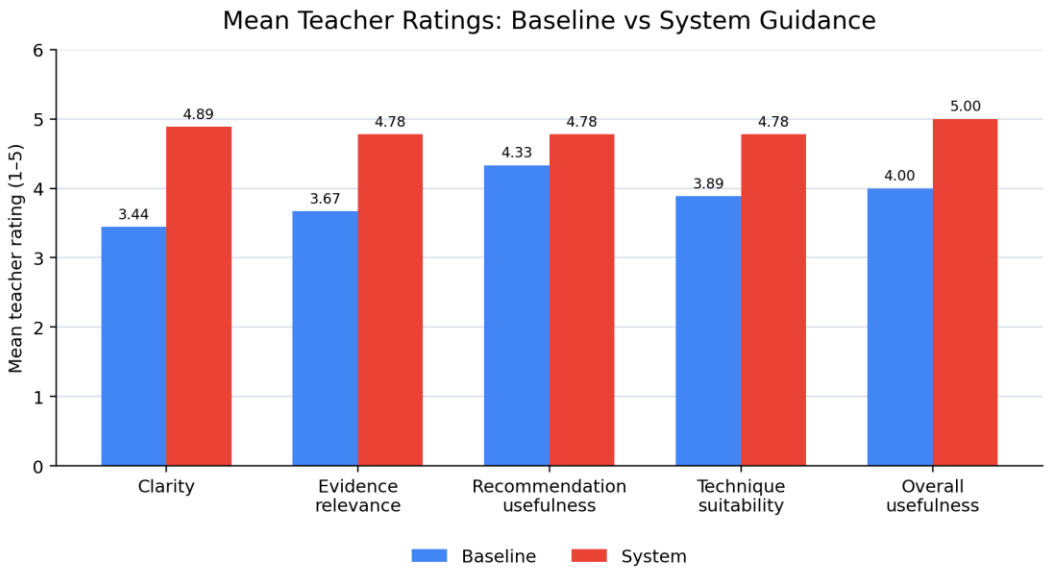


Figure 6.1. Mean teacher ratings for baseline and system guidance across five criteria.

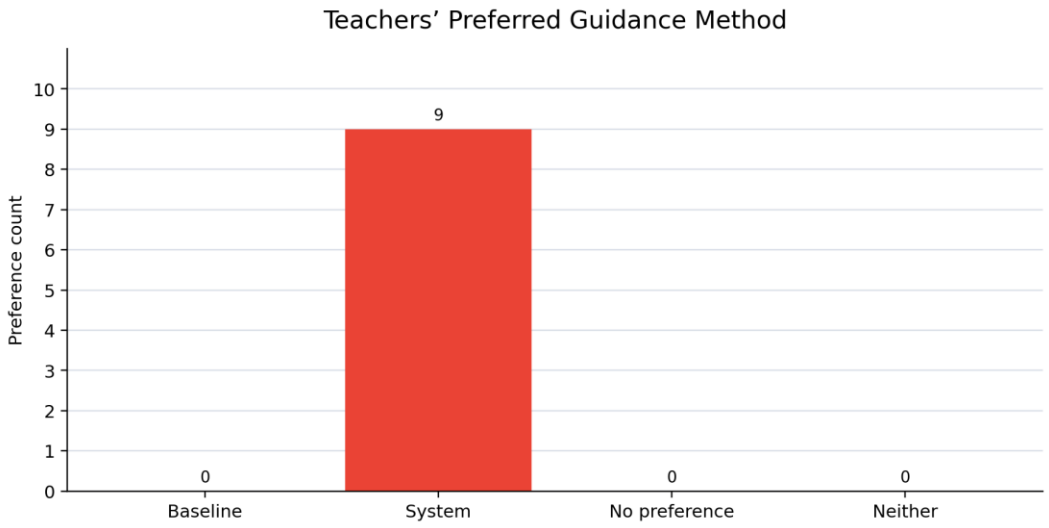


Figure 6.2. Teacher preference counts across nine scenario comparisons.

System guidance was preferred in 9/9 scenario comparisons; baseline, no-preference, and neither counts were zero. These are descriptive results from three evaluators and three scenarios, not independent evidence from nine different teachers. They do not establish statistical significance or generalisability.

6.2 Research results: adaptive-MCQ expert evaluation

The MCQ evaluation workbook contains three teacher responses for three lesson packs. Adaptive MCQs had higher descriptive means than baseline MCQs on all reported suitability criteria.

Table 6.2. Preliminary adaptive-MCQ ratings (three teachers)

Criterion	Baseline mean (/5)	Adaptive mean (/5)	Difference
Wording manageability	1.83	4.53	+2.70
Difficulty suitability	2.00	4.27	+2.27
Directness	2.03	4.23	+2.20
Overall suitability	2.33	4.33	+2.00

Figure 6.3 compares mean teacher ratings for the two MCQ sets across the four reported criteria.

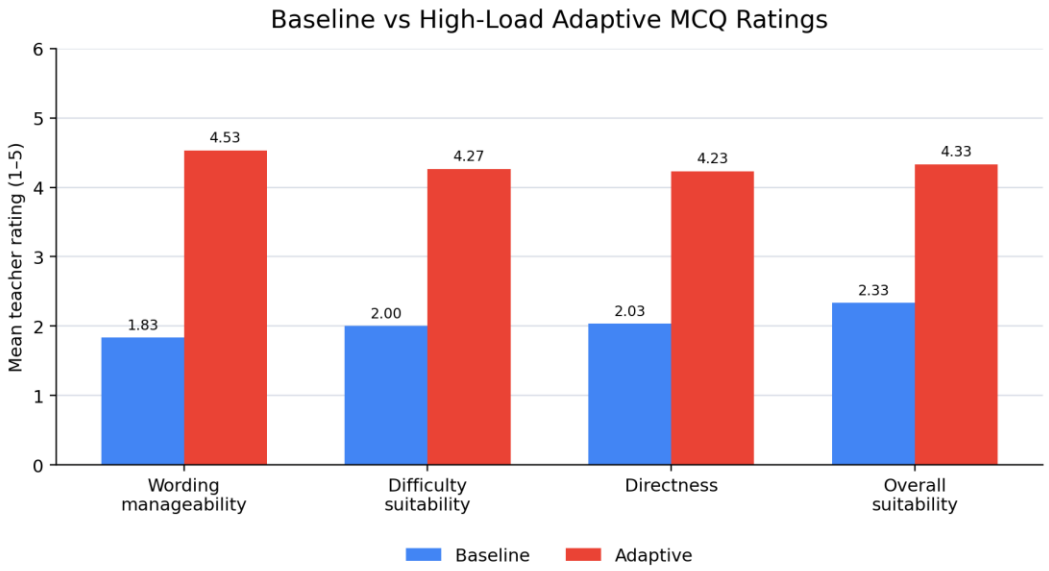


Figure 6.3. Mean teacher ratings for baseline and high-load adaptive MCQs across four criteria.

All three teachers preferred the adaptive set. Answer-key review recorded 30/30 affirmative judgments for baseline items and 29/30 for adaptive items. One adaptive judgment was negative and did not include an explanatory note; it is therefore an item requiring review, not a confirmed error. These findings indicate preliminary expert perception, not verified correctness for all generated questions or evidence of student learning benefit.

Figure 6.4 presents the answer-key judgements: one adaptive item was rated “No,” while the remaining judgements were “Yes”; none were marked “Uncertain.”

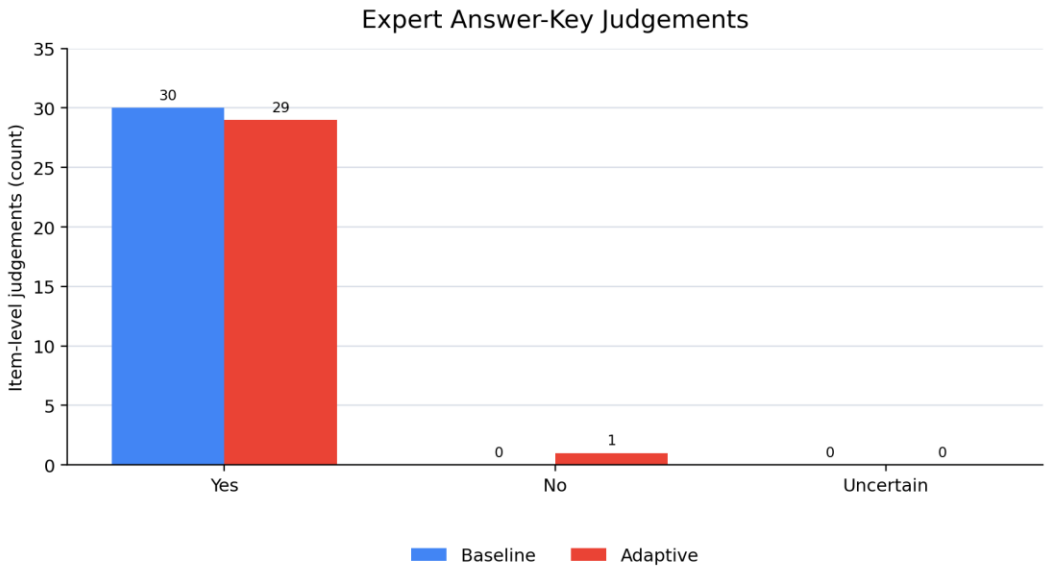


Figure 6.4. Expert answer-key judgements for baseline and adaptive MCQs (30 item judgements per method).

6.3 Product and technical results

The project test report records 39/39 automated tests passing across the four reported groups. Separately, the cognitive-style service test suite passed 10/10 tests during thesis preparation. Those tests cover style-specific aggregation, prompt construction/translation, cached profile behaviour, MongoDB synchronization, and model-artifact integration. The tests exercise defined code paths, with external generation mocked where applicable; they do not measure style-prediction accuracy or independently assess whether teachers find generated explanations faithful and useful. No live model robustness or predictive-accuracy result is inferred from these pass rates.

6.4 User or stakeholder validation: exploratory classroom worksheet

The current user-testing workbook contains ten populated learner records for the INTERNET lesson. Each learner answered ten questions. The recorded correct counts were 8, 7, 9, 9, 8, 7, 8, 7, 7, and 8, totalling 78 correct answers out of 100 (78.0%). The mean individual score was 7.8/10.

The recorded cognitive-load labels were Low (1 learner), Medium (2), High (6), and Very High (1). Ten clarity ratings were present: 3, 4, 5, 5, 4, 4, 5, 4, 4, and 5, with a

mean of 4.30/5. The workbook contains no perceived-difficulty ratings, so perceived difficulty is not analysed.

One teacher-review row is present, completed by the single teacher who reviewed this classroom test. It records that all questions matched the lesson content, all answer keys were correct, the difficulty was very appropriate, all wording was clear, class understanding was good, the next-lesson recommendation was appropriate, and overall usefulness was rated “Useful.” No item-specific correction or comment is entered. This is one teacher's judgement and cannot establish a general teacher consensus or independently prove correctness.

Table 6.3. Exploratory classroom worksheet summary

Measure	Recorded result
Session date	3 October 2026
Lesson	INTERNET
Learners	10 Grade 8 students
Teacher reviewers in classroom session	1
Populated learner rows	10
MCQs per learner	10
Correct responses	78/100
Aggregate score	78.0%
Mean learner score	7.8/10
Mean recorded clarity rating	4.30/5 (n = 10)
Perceived-difficulty responses	Not recorded
Completed teacher-review rows	1

These scores describe performance on a single post-lesson quiz. There is no pre-test or comparison condition in this worksheet; accordingly, the scores cannot demonstrate that the platform improved learning or that the generated MCQs caused the observed performance. The teacher review provides a separate, preliminary judgement of content alignment, answer key, difficulty, clarity, and next-lesson recommendation.

6.5 Cognitive-style explanation validation boundary

The repository contains a separate cognitive-style explainability service and its own automated tests. The 10/10 passing tests provide functional evidence for selected operations, including combining LIME/SHAP attribution magnitudes, preserving attribution direction, preparing a teacher-oriented prompt, handling the Intermediate

class label, and reusing or refreshing saved analysis. The available teacher evaluation workbook does not contain a separate rating task for cognitive-style explanations. Therefore, no teacher preference, clarity score, factual-fidelity result, or learner outcome is claimed for this output. A focused teacher review across all three predicted classes and representative evidence patterns remains future evaluation work.

6.6 Summary of findings

The documented project tests and separate cognitive-style suite passed for their specified behaviours. Three teachers gave favourable descriptive ratings to load-related system guidance and adaptive MCQs compared with the supplied baselines. The cognitive-style explanation itself did not receive a separate teacher-quality evaluation. Separately, the classroom worksheet records results from ten learners and one teacher review. No perceived-difficulty data are recorded, and session usability observations are absent. The results support a preliminary pilot description, not a claim of educational effectiveness.

7 DISCUSSION

7.1 Interpretation of findings

The guidance evaluation suggests that the teachers in the recorded comparison generally preferred the system outputs on clarity, evidence relevance, recommendations, and technique suitability. The adaptive-MCQ evaluation shows a sizeable descriptive rating difference in favour of adaptive sets. Because the sample consists of three teachers and the evaluations cover a small number of scenarios and lesson packs, these differences should be treated as early feedback to guide refinement rather than as population estimates.

The classroom worksheet demonstrates that the platform workflow and quiz were used with ten learner records. The mean score and clarity rating are useful descriptive observations. They cannot show that the system caused learning gains: no baseline quiz, comparison group, or pre/post design is recorded. The one teacher's positive review is useful as a case-level quality check, but it is not enough to validate every generated item or recommendation across contexts.

7.2 Comparison with literature, baselines, and existing solutions

The component follows literature-motivated practices by separating explanation from intervention, using retrieval questions, and preserving human oversight around generated educational content. The teacher A/B results compare the system with the baseline used in the project evaluation. The available evaluation artefacts document the paired ratings and method-masking procedure but provide limited detail about how each baseline response was authored; this constrains interpretation of the comparison. The adaptive MCQ comparison is particularly relevant to perceived wording and difficulty, while the answer-key review exposes why comparative preference and factual correctness must be measured separately.

7.3 Research contribution

The research contribution includes two explicit XAI-to-language workflows: magnitude-ranked signed LIME/SHAP evidence for explaining cognitive-style predictions, and a positive-only LIME/SHAP support list for load-related guidance.

It also documents a class-level verification workflow and a separate lecture-grounded assessment path. The contribution is not a new attribution theory, a validated cognitive-load or cognitive-style instrument, or proof that generated support improves attainment.

7.4 Product contribution

The product contribution is a connected workflow that can generate cognitive-style and cognitive-load explanations, learner actions, next-lesson recommendations, and adaptive MCQs using task-specific inputs and validation. Teacher review applies to individual guidance and next-lesson planning. MCQ generation has structural validation and can be reviewed by a subject teacher in a study procedure, but the service does not enforce mandatory teacher approval before every learner quiz. This distinction must remain explicit in product claims.

7.5 Limitations and threats to validity

Construct validity is limited because behavioural traces are proxies for cognitive load, and a model category is not a direct subjective measurement. Internal validity is limited for the classroom results because there is no pre-test or comparison condition. External validity is limited by the small number of teachers, lesson packs, and learner records. The classroom worksheet has ten learner entries but only one teacher review. Learners provided a 1–5 rating about their understanding of the MCQs, but the exact scale anchors were not recorded; no written comments or separate perceived-difficulty ratings were entered. Selection, subject familiarity, teacher support, and lesson-specific difficulty may influence scores.

The A/B studies are descriptive and small. A preference can be influenced by presentation, baseline quality, or evaluator expectations despite concealed method labels. The cognitive-style explanation has no separate teacher-rating or explanation-faithfulness study in the available records. Structural validation cannot catch all semantic MCQ errors. CoVe checks an evidence snapshot and may not detect omissions or errors in that snapshot. Prompt constraints do not guarantee perfect language quality. These limitations prohibit causal, statistical-significance, and broad generalisability claims.

7.6 Implications

For teachers, generated explanations and recommendations should be presented as decision support, with underlying evidence and review controls. For learners, quiz scores should be used as feedback about the current attempt, not as fixed ability judgements. For developers, the result supports maintaining clear distinctions between model prediction, local attribution, generated text, deterministic validation, and human review. Further evaluation should include complete session records and more systematic item-level content review.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The component's potential value proposition is to help teachers interpret online-learning signals, prepare learner actions and next-lesson plans, and draft lesson-grounded quizzes. Likely users include schools and training providers already using digital lessons and teachers responsible for reviewing learning materials. The present evaluation does not establish willingness to pay, market demand, procurement readiness, or cost savings.

8.2 Market and competitive position

The component combines learning analytics, explainability, teacher workflow, and generative assessment support. It may be differentiated from tools that provide only dashboards or generic content generation by maintaining evidence links and human-review states. A defensible comparison requires a current market scan and feature-by-feature evaluation, which are not included in the available project artefacts; no unsupported competitor or market-size claim is made here.

8.3 Business or sustainability model

Potential sustainability options include institutional licensing, a hosted subscription for schools, or deployment and support agreements. Operating costs would include cloud hosting, database and storage, language-model calls, security maintenance, technical support, and content review. A pilot should measure usage and operating cost before selecting a pricing model. The software should retain a usable fallback for core workflows where possible so that model-provider outages do not remove all support functionality.

8.4 Scalability, intellectual property, and compliance

Scaling requires load testing, role-based access, privacy-by-design, reliable audit records, retention and deletion policies, backup and incident response, and model/provider version tracking. Ownership of student work, project code, third-party libraries, generated content, and branding is a deployment-readiness matter for the institution and project team; it was not measured in this evaluation.

8.5 Contribution to the Sustainable Development Goals

The project is most directly aligned with Sustainable Development Goal 4 (Quality Education) through its aim to support accessible and responsive learning. This is an intended alignment, not evidence that the software has improved educational access or outcomes. Inclusive design, accessibility testing, language support, device availability, and teacher training would be needed to substantiate broader impact.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presents an individual learning-support contribution with two distinct teacher-facing explanations: a cognitive-style explanation based on magnitude-ranked, signed LIME/SHAP evidence and a cognitive-load explanation based on positive signals supporting elevated load. It also covers learner recommendations, next-lesson guidance, and adaptive MCQs. Structured generation, evidence checks, output validation, teacher review, and deterministic fallback provide safeguards appropriate to the different workflows. The available evidence includes 39 documented project-wide tests, a separately run 10-test cognitive-style service suite, preliminary blinded comparisons involving three teachers, and a classroom pilot with ten learners and one teacher reviewer. The cognitive-style explanation has not been separately evaluated by teachers. The descriptive evidence does not establish learning improvement, general usability, style-explanation quality, or universal MCQ correctness.

9.2 Achievement of objectives

The repository documents implementation and tests for cognitive-style explanation, positive-signal load aggregation, structured teacher/learner guidance, evidence-based next-lesson planning, and lecture-grounded MCQ generation. The teacher A/B workbooks provide preliminary comparative ratings for load-related guidance and MCQs, but not for style explanation. The classroom workbook contributes a small real-use description for one lesson. Objectives concerning style-explanation quality, broad usability, learning effectiveness, and cross-context reliability remain open.

9.3 Individual research and product contributions

The individual research contribution is the design and evaluation framing for connecting local cognitive-style and cognitive-load explanations to reviewable educational support. The product contribution is implementation and integration of the style explanation, load explanation and recommendations, next-lesson recommendation, and adaptive MCQ outputs with distinct evidence boundaries and

validation rules. Final contribution attribution must match the team's approved contribution statement.

9.4 Future directions

Conduct a focused teacher evaluation of the cognitive-style explanation across Visual, Verbal, and Intermediate outputs, checking clarity, whether cited behaviours match the displayed technical evidence, whether the explanation avoids fixed-trait or causal claims, and whether its boundaries are understood. Preserve the exact user-test script and session procedure. Add a short, consistently administered learner feedback instrument if perceived difficulty or usability is a research outcome. Conduct additional subject-teacher item reviews, including reasons for ambiguous or incorrect items. Evaluate task completion and interaction issues for both teacher explanations, learner recommendations, next-lesson planning, and quizzes. A separate controlled study with suitable pre/post or comparison measures would be needed to investigate learning effects. Improve deterministic language checks, test across subjects and lesson formats, and evaluate accessibility, fairness, and reliability.

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GLOSSARY

Adaptive MCQ: A question generated with its wording or task demand adjusted using a learner or class cognitive-load category. **Cognitive-load estimate:** A model-produced category inferred from interaction data; not a direct psychological measurement. **Cognitive-style classification:** A model-produced Visual, Verbal, or Intermediate category interpreted as contextual evidence, not a permanent learner trait. **Evidence snapshot:** Structured aggregate data supplied to the next-lesson recommendation process. **Local attribution:** A feature contribution associated with a particular model prediction. **Positive-only signal list:** A ranked list that includes only positive LIME/SHAP contributions under the component's specified policy.

APPENDICES

Appendix A. Classroom user-testing instruments

This appendix reproduces the blank fields and teacher questions used for the exploratory classroom test. The accompanying data-collection workbook is `USER_TESTING_DATA_COLLECTION.xlsx`. The completed participant-level rows are not reproduced here; the aggregate results are reported in Chapter 6.

A.1 Student result record (one row per learner)

Field	Entry / response
Anonymous participant ID	S01–S10; do not enter a learner name
Lesson	Lesson code/title
System cognitive-load result	Record the category exactly as displayed
Total MCQs	Number of questions attempted
Correct answers	Number correct
Score (%)	Correct answers ÷ total questions × 100
MCQ understanding / clarity	Learner selects 1–5 after being asked whether they understood the MCQs; exact wording and scale-anchor labels were not recorded
Perceived difficulty (optional)	Learner rating, 1–5, only if asked; not populated in the current dataset
Comments / observed issues	Short note, if collected

A.2 Teacher review form (one response for the classroom session)

1. Do the MCQ questions match the content of the lesson? (All / Most / Some / No)
2. Is the answer key correct? (All answers / Most answers / Only a few / No)
3. Is the difficulty level appropriate for the students? (Very appropriate / Appropriate / Somewhat difficult / Somewhat easy)
4. Are the questions written clearly and easy to understand? (All / Most / Some / No)
5. Based on the students' results, what level of understanding does the class appear to have? (Good / Moderate / Limited / Not enough information to judge)
6. Is the system's next-lesson recommendation appropriate? (Yes / Partly / No)
7. Overall, how useful are the generated MCQs and next-lesson recommendation? (Very useful / Useful / Slightly useful / Not useful)
8. Please note question numbers, corrections, explanations, or other comments.

After the quiz, learners were asked to rate whether they understood the MCQs on a 1–5 scale and were offered an optional comment field. No comments were submitted. The exact rating anchors were not retained in the worksheet.

The classroom evaluation workbook contains one teacher response. It is not the same study as the separate three-teacher blinded A/B comparisons described in Chapter 6.

Appendix B. Teacher-guidance A/B evaluation instrument

The `Teacher_Guidance_Evaluation_Automated_Template_FIXED.xlsx` workbook contains the data-entry structure for the paired comparison. Each teacher rated both methods for each scenario on a five-point scale. The participant-facing method labels were Set A and Set B; the private mapping was held separately during the blind evaluation and is not reproduced in this instrument.

B.1 Rating fields recorded for each participant, scenario, and set

Rating criterion	Recorded value
Clarity	1–5
Evidence relevance	1–5
Recommendation usefulness	1–5
Technique suitability	1–5
Overall usefulness	1–5

B.2 Preference fields

For each scenario, the evaluator selected Set A, Set B, No preference, or Neither. An optional comment could explain the preference. The workbook stores participant code, scenario, selected set, resolved preferred method, and comment. Three teachers evaluated three scenarios, giving nine paired scenario comparisons.

Appendix C. Adaptive-MCQ expert evaluation instrument

The MCQ expert-evaluation form compares Set A and Set B independently for one lesson pack. The evaluator records an anonymous participant code, lesson code, subject/specialisation, teaching experience, and familiarity with the lesson topic. The baseline/adaptive mapping is not shown to the evaluator during the blind comparison.

For **each question in each set** (ten questions per set), the evaluator records:

Criterion	Scale anchors
Wording manageability for a high cognitive-load learner	1 = Very difficult to process; 5 = Very manageable
Difficulty suitability for a high cognitive-load learner	1 = Highly inappropriate; 5 = Highly appropriate
Directness and focus	1 = Not direct; 5 = Very direct
System-selected answer is correct and the only valid answer	Yes / No / Uncertain

The evaluator also gives an overall suitability rating for each set, selects which set is more suitable for learners experiencing high cognitive load, and may provide comments. The form asks the evaluator to list any incorrect, ambiguous, or unsupported questions and explain the issue. The three responses are summarised in Chapter 6; individual responses remain in the evaluation workbook.

Appendix D. Extended evaluation results

The anonymised aggregate tables in Chapter 6 report the classroom scores, teacher-guidance criterion means and preferences, adaptive-MCQ criterion means, and answer-key review counts. The completed Excel workbooks are retained as research artefacts. Do not include names, direct identifiers, or the private A/B mapping in a public-facing appendix.

Appendix E. Academic integrity and AI-use disclosure

[List the tools used, their purposes, representative prompts or workflows where required, how outputs were verified, and which sections or artefacts were affected, following the current module policy.]

Appendix F. Anonymised classroom pilot records

The following table transcribes the learner-level values in `USER_TESTING_DATA_COLLECTION.xlsx`. Participant codes are retained for row matching; names and direct identifiers are not included. The percentage is the recorded correct-answer count divided by ten questions. These rows describe one post-lesson quiz and are not a pre/post comparison.

Participant	Load category shown	Correct / 10	Score (%)	MCQ clarity (1–5)
S01	Low	8	80	3
S02	High	7	70	4
S03	High	9	90	5
S04	Medium	9	90	5
S05	Very High	8	80	4
S06	High	7	70	4
S07	High	8	80	5
S08	High	7	70	4
S09	Medium	7	70	4
S10	High	8	80	5
Total / mean	—	**78 / 100**	**78.0%**	**4.30 / 5**

The workbook contains one teacher-review row for this lesson. The recorded answers were: MCQs matched lesson content—All; answer key correct—All; difficulty—Very appropriate; wording—All of them; class understanding—Good; next-lesson recommendation—Yes; overall usefulness—Useful. The optional correction/comment field was empty. This is one teacher's recorded judgement, not a multi-teacher consensus.

Appendix G. Baseline teacher-guidance ratings

The table gives the nine baseline rows recorded in `Teacher_Guidance_Evaluation_Automated_Template_FIXED.xlsx`. T-codes and scenario codes are anonymous identifiers. The five values are the original 1–5 ratings, not recalculated averages.

Teacher	Scenario	Clarity	Evidence relevance	Recommendation usefulness	Technique suitability	Overall usefulness
T001	S01	4	4	5	4	4
T001	S02	3	3	4	4	4
T001	S03	3	4	4	4	4
T002	S01	4	4	4	3	4
T002	S02	3	4	5	5	4
T002	S03	4	4	5	4	4
T003	S01	3	4	3	3	4
T003	S02	4	3	5	4	4
T003	S03	3	3	4	4	4

Appendix H. System teacher-guidance ratings

The paired system rows from the same workbook are reproduced below using the same criteria and scale as Appendix G. Each teacher rated both methods for each of three scenarios; the nine rows are therefore paired scenario judgements, not nine independent teachers.

Teacher	Scenario	Clarity	Evidence relevance	Recommendation usefulness	Technique suitability	Overall usefulness
T001	S01	5	5	5	5	5
T001	S02	5	5	4	4	5
T001	S03	5	5	4	4	5
T002	S01	5	4	5	5	5
T002	S02	5	5	5	5	5
T002	S03	5	5	5	5	5
T003	S01	5	5	5	5	5
T003	S02	5	5	5	5	5
T003	S03	4	4	5	5	5

Appendix I. Adaptive-MCQ expert rating summary

The following anonymised teacher-by-set summaries are transcribed from the `Clean_Data` sheet in `MCQ Expert Evaluation Responses (1).xlsx`. Rating means cover ten questions per set. Answer-key judgements count whether the system-selected answer was correct and the only valid answer.

Teacher	Method	Wording mean	Difficulty mean	Directness mean	Overall suitability	Yes / No / Uncertain
T001	Baseline	1.8	1.8	1.9	2	10 / 0 / 0
T001	Adaptive	4.4	4.0	4.3	4	10 / 0 / 0
T002	Adaptive	4.5	4.4	4.1	5	9 / 1 / 0
T002	Baseline	1.7	1.7	2.1	3	10 / 0 / 0
T003	Baseline	2.0	2.5	2.1	2	10 / 0 / 0
T003	Adaptive	4.7	4.4	4.3	4	10 / 0 / 0

These are descriptive summaries of three evaluators and three lesson packs. The one negative adaptive answer-key judgement remains an item requiring review; it is not independently confirmed as an incorrect item because no explanation was recorded.

Appendix J. Functional test cases FT01–FT10

This selected test-case catalogue is from the detailed 20-test report for the LIME–SHAP/guidance services. It is not a replacement for the separate project-wide summary of 39 automated tests in Chapter 6. The recorded report used Python `unittest`, mocked Gemini responses, and in-memory SQLite for persistence tests; all listed cases were reported as passing.

ID	Test case	Verification purpose	Result
FT01	Normalize and rank aggregate signals	Unique positive features return in expected Top-3 order	Pass
FT02	Normalize positive sums and average	LIME and SHAP normalize independently before averaging	Pass
FT03	Remove zero and negative values	Non-positive contributions are excluded from selected signals	Pass
FT04	Handle all non-positive inputs	Empty result is returned when no positive support exists	Pass
FT05	Deterministic tie handling	Tied features are stable when input order changes	Pass
FT06	Save and update Top-3 signals	Existing student–lesson record updates without duplication	Pass
FT07	Load complete cached analysis	Complete saved analysis avoids unnecessary regeneration	Pass
FT08	Refresh outdated guidance	Guidance refresh does not rerun saved LIME/SHAP explanations	Pass
FT09	Reject incomplete cache	Top-signals-only data is not treated as complete analysis	Pass
FT10	Share saved guidance	Teacher-approved guidance is available to the selected student	Pass

Appendix K. Functional test cases FT11–FT20

The remaining cases from the same detailed report are shown below. A pass indicates that the predefined scenario met its expected behaviour; it is not a measure of predictive accuracy or educational effectiveness.

ID	Test case	Verification purpose	Result
FT11	Save feedback only after sharing	Feedback is accepted only for previously shared guidance	Pass
FT12	Reject and withhold guidance	Rejected guidance is withheld from the student	Pass
FT13	Regenerate and archive feedback	Regeneration resets review state and preserves feedback history	Pass
FT14	Calculate dominant cognitive load	Prediction counts and dominant load are calculated for a student/lesson	Pass
FT15	Batch local-model prediction	Multiple rows are predicted with API-compatible rounding	Pass
FT16	Reject invented matched behaviour	Guidance cannot cite behaviour absent from supplied evidence	Pass
FT17	Generate guidance with one AI call	One mocked call generates and validates the constrained guidance bundle	Pass
FT18	Reject malformed JSON	Invalid model response follows controlled error handling	Pass
FT19	Reject unsupported technique	Suggestions outside the constrained catalogue are rejected	Pass
FT20	Enrich cached technique locally	Cached techniques gain learner details without another AI call	Pass

Appendix L. Evaluation calculation notes

This appendix makes the arithmetic behind the reported descriptive metrics explicit. It does not introduce a new outcome or imply causation.

Measure	Calculation from recorded data	Reported value
Classroom quiz total	8 + 7 + 9 + 9 + 8 + 7 + 8 + 7 + 7 + 8 correct answers	78 / 100
Classroom aggregate percentage	78 correct ÷ 100 attempted × 100	78.0%
Mean learner score	78 correct ÷ 10 learners	7.8 / 10
Mean MCQ clarity rating	(3 + 4 + 5 + 5 + 4 + 4 + 5 + 4 + 4 + 5) ÷ 10	4.30 / 5
Classroom load-category counts	Count the displayed category in each of the ten learner rows	Low 1; Medium 2; High 6; Very High 1
Teacher-guidance paired cases	3 teachers × 3 scenarios, with both methods rated in each case	9 paired cases; 9/9 preferred System
Guidance ratings per criterion and method	One rating from each teacher for each of three scenarios	n = 9 per method and criterion
MCQ answer-key judgements	3 teachers × 10 items for each method	30 judgements per method
Baseline answer-key positive rate	30 Yes ÷ 30 judgements × 100	100.0%
Adaptive answer-key positive rate	29 Yes ÷ 30 judgements × 100	96.7%; 1 No, 0 Uncertain

The classroom score is a single post-lesson quiz result, not a measured gain. Teacher ratings and answer-key counts are descriptive summaries of the recorded evaluators and items; they are not inferential statistical tests.